



Evaluation of INSAT-3DR sounder derived wind index using radiosonde observation over India

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Received: 21 February 2024 / Accepted: 25 October 2024 / Published online: 27 December 2024
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Abstract

Rapid advancements in the field of forecasting demand in-depth study across various climate fields, including wind-derived products. The wind index is a highly valuable parameter that exhibits significant variability and intermittence across various timescales due to its dependence on weather conditions. Therefore, accurate wind index observation is acknowledged as a vital contribution to trustworthy large-scale Thunderstorm forecasts. Nevertheless, the satellite-derived wind index observations are often associated with errors and require extensive evaluation before their use in operational forecasts. The present study aims to assess the wind index retrieved from the Infrared (IR) sounder data from the Indian National Satellite System (INSAT)-3DR using in-situ radiosonde data. It was found that the results closely resemble the unfavourable skew of the satellite data. The assessment shown in the present study is carried out for years 2020, 2021 and 2022 or different stations spread over vast India. Furthermore, this paper emphasizes the Wind Index accuracy for helicopter operation. However, wind index information is derived by satellite but always missing by Radiosonde. The proposed method is applied to the estimating Radiosonde wind index. This study mainly focuses on calculating wind index from different radiosonde parameters and then validating INSAT 3DR with Radiosonde WI.

1 Introduction

Thunderstorms are one of the most disastrous natural phenomena. It is an atmospheric phenomenon that helps dissipate the vertical instabilities of the atmosphere that cause heavy rainfall, lightning, and other extreme weather phenomenon such as downbursts (Federico et al. 2020). A downburst is a natural weather event wherein a strong downdraft of air rushes toward the surface and results in a strong outflow of straight-line winds at or just above the ground. Though all thunderstorms have downbursts but sometimes these are so powerful that it results in havoc.

Before 1970, there was not much knowledge of downbursts. The damages done due to the downbursts were considered due to tornadoes. It was Dr. Ted Fujita who first

time noticed that damages in the aftermath of a "super-outbreak" of 148 tornadoes on April 3–4, 1974 were from some weather events other than tornadoes. Fujita (1985) enunciated the concept of a downburst and categorized it into two parts as per their sizes. The downbursts larger than the size of 4 km spatial extent are termed as macro bursts and the smaller ones are termed as microbursts (Djordje et al. 2022). The microbursts are further subdivided into two classes: dry microbursts and wet microbursts. The dry microburst is a microburst that concurs with lesser surface rainfall (Gary 1989). Winds may reach up to 60 m/s for intense downburst cases. As a matter of fact, wind is one of the most important parameters that causes the most aviation accidents and delays (Gultepe et al. 2019). The high winds in the form of downbursts are very fatal for aviation. Downbursts pose a risk to aircraft in flight because of the strong wind shear they create, especially during take-off and landing phases (Wolfson 1990, Wakimoto 1985, Jacob et al. 2022). High-resolution Doppler radars are not only capable to detect microbursts but also provide some signatures that help in locating them (Roberts and Wilson 1989; Hermes et al. 1993). Now-casting or the forecasting of downbursts is essential for saving the losses of lives and property. Many attempts have been made to accurately predict the events

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of downbursts. Donald W. McCann (1994) proposed a new index which he termed WINDEX for predicting microbursts. The methodology to compute Windex is based on the identification of the air masses that can eventually turned into microbursts. The atmospheric soundings from radiosonde observations (RAOBs), space-based observations, or model forecasts provide the knowledge of air masses required for Windex computation (Houchi, et al. 2009).

Several studies have been carried out to exploit GOES sounder observations for the computation of wind index and wind gusts. The geostationary soundings from the GOES satellite also helped in assessing the diurnal performance of Windex. (Kenneth and Ellrod 2004). The formation of the nocturnal temperature inversion after sunset causes the WINDEX to drop during the evening hours since WINDEX is dependent on the square of the temperature lapse rate (Ellrod et al. 2000, 1990). In addition to WINDEX's dependence on lapse rate, Pryor et al. (2004) pointed out that the size of the convective available potential energy (CAPE) pool that can "fuel" nocturnal convective systems may have an impact on the strength of convective wind gusts (Suomi et al. 2013, Stoffelen et al. 2019) The WINDEX algorithm does not specifically take CAPE as its input parameter.

INSAT-3DR satellite launched in 2016. Since then, it has been providing temperature and humidity soundings from geostationary orbit using its 18 infrared sounder channels over data sparse Indian ocean region. Many meteorological geophysical parameters are operationally retrieved and used by the India Meteorological Department (IMD) to support its weather services. WINDEX is also one of the indices that are derived using INSAT-3DR sounder observations. It is always necessary to evaluate various meteorological parameters derived from satellite to estimate its uncertainties and accuracies. It helps in improving the estimation methodologies and for its best use for routine weather now-casting and forecasting applications. This work is aimed towards evaluating the performance of INSAT-3DR derived WINDEX with WINDEX computed using RAOBs soundings over the Indian region.

2 Data and methodology

This work presents an inter-comparison of Windex computed from RAOBs soundings and INSAT-3DR Windex product. Thus, this study uses INSAT-3DR and radiosonde data. The radiosonde (RS) and rawinsonde (RW) network in India is operated and maintained by the IMD. IMD's present upper air observational network comprises 39 radiosondes and 62 pilot balloon observatories spread all over the country. Generally, the RS and the pilot balloons are launched twice a day at 0 and 12 UTC. However, due to operational difficulties and operational costs sometimes only 0 GMT

observations are taken. These RS/RW measurements provide the vertical profiles of temperature, humidity, and winds. For this study, only a limited number of stations and only 0 GMT observations are taken. The selection of stations is done based on consistency in the form of data set availability. Only that data is considered to be good where there are vertical profiles of temperature and moisture are available at least up to 100 hPa (WMO 2014). It has also been found that observations are more sporadic at 12 UTC, therefore only 0 UTC observations are considered for the present work. The RAOBs are taken from the Department of the Atmospheric Science University of Wyoming (<http://www.uwyo.edu/atasc>). Figure 1 shows the total number of stations considered for analysis.

As it has already been stated the INSAT-3DR satellite is in geostationary orbit at 74°E longitude. It provides hourly soundings over the Indian land mass, Bay of Bengal, and Arabian Sea. In every six hourly intervals, the observations are taken over the Indian Ocean region. The calibrated, navigated radiances/brightness temperature observations are provided in the level-1B file whereas the retrieved temperature, moisture profiles, and other derived products are available in level-2B files (Giri et al. 2023). All data products of INSAT-3DR sounder are available in hierarchical data format-5 (HDF-5) format. The link <https://www.mosdac.gov.in> provides a detailed description of various data products and the formats of INSAT-3DR. A detailed description of the retrieval methodology of INSAT-3DR imager and sounder products is provided at https://www.mosdac.gov.in/docs/INSAT_3D_ATBD_MAY_2015.pdf

For the present study, the three years of data from 2020 to 2022 of RAOBs and Level 2B (L2B) products of INSAT-3DR is taken. To carry out this study, first of all, a quality check is applied to RAOB data. Any RAOB soundings that show the temperature inversion at lower levels or, do not have measurements up to melting point level or if all temperature values recorded are greater than 0.0°C is rejected. The quality checks are not defining the quality of the data set but are devised to satisfy the Eq. (1) well. The quality filtering step is followed by the spatial and temporal collocation of The RAOBs and INSAT-3DR soundings procedure. Before temporal collocation, it was necessary to reduce the volume of the INSAT-3DR sounding data. To do this only those files of INSAT-3DR are selected for which RAOB observations are available. The threshold of ± 30 min is used for temporal collocation. For spatial collocation, the latitude and longitude of a given RAOB is searched within a pixel of INSAT-3DR sounder. Since nominal resolution of INSAT-3DR sounder is 10 km at nadir therefore, the collocation criterion is approximately 10 km. The temperature and moisture soundings of RAOBs and INSAT-3DR soundings are further processed to compute Windex from both.

Fig. 1 Flow chart for computing Windex using temperature and humidity profile

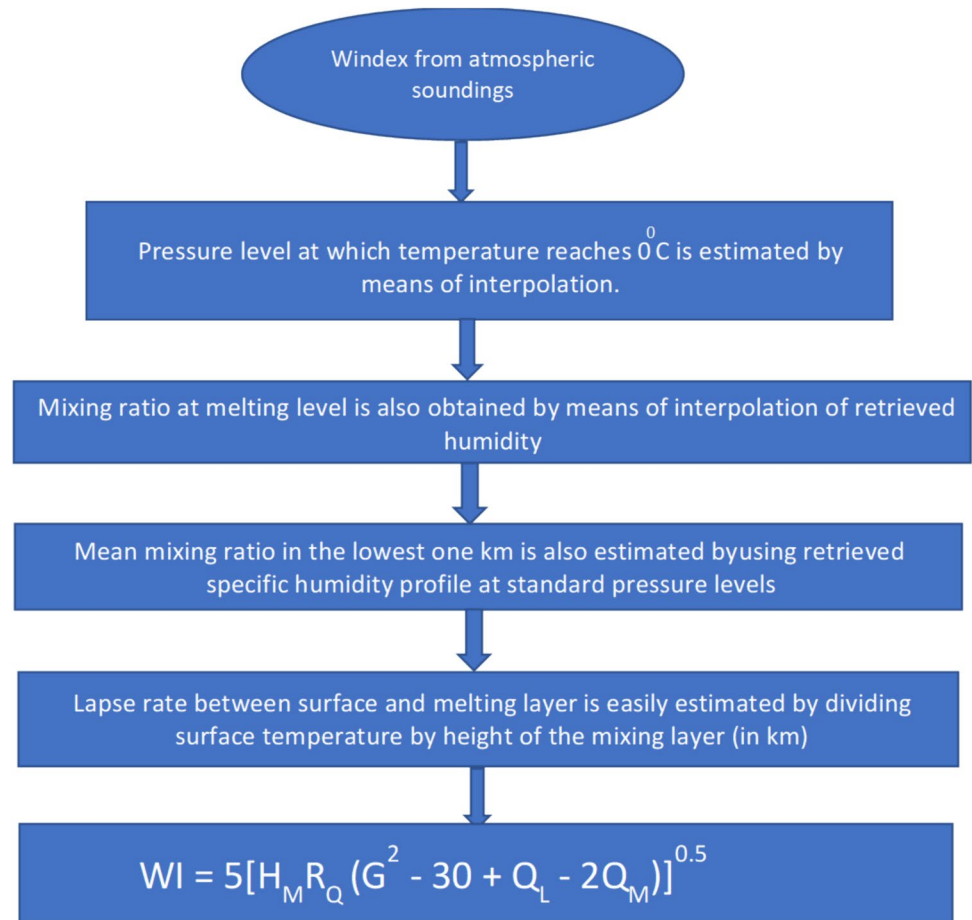


Table 1 Abbreviation of Windex

H_M	Melting Levels (height in kilometres above ground)
G	lapse rate (degrees Celsius per kilometres)
Q_L	The Mixing Ratio (1 km above the surface)
Q_M	The mixing ratio at the melting level
R_Q	$Q_L/12$

The formula for computing Windex from temperature and moisture profile is given in Eq. (1)

$$WI = 5[H_M R_Q (G^2 - 30 + Q_L - 2Q_M)]^{0.5}$$

The steps of computation of Windex from atmospheric soundings are as follows Abbreviation of Windex is mentioned in Table 1.

- 1) From the retrieved temperature profile at standard pressure level, the pressure level at which temperature reaches 0.°C is estimated by means of interpolation.

Height of the melting layer (Table 1) is then estimated by using the standard formula based on hydrostatic approximation. (INSAT ATBD, 2007)

- 2) Mixing ratio at melting level is also obtained by means of interpolation of retrieved humidity.
- 3) Mean mixing ratio in the lowest one km is also estimated by using retrieved specific humidity profile at standard pressure levels.
- 4) Lapse rate G between surface and melting layer is easily estimated by dividing surface temperature by height of the mixing layer (in km).

These parameters are then substituted in the formula for WI given in Eq. (1).

The procedure to compute Windex is summarised in flow chart shown in Fig. 1.

After following these steps, a collocated match-up data of Windex of RAOB and INSAT-3DR soundings is generated. The number of match-up data points from the year 2020 to 2022 are 985, 618 and 1129 respectively. These match-up data points are then used to estimate the error budget in Windex in terms of root mean square error (RMSE), correlation,

maximum error (ME), and percentage bias. Additionally, Windex is classified into four categories as per its potential to generate wind gusts. The four categories of Windex are defined by the range as (i) $WI < 10$ (ii) $10 \leq WI < 49$ (iii) $49 \leq WI < 79$ and (iv) $WI \geq 79$.

This categorization is simply done on the Wet Microbursts Severity Index (WMSI) values. The WMSI is taken as a reference because Pryor et al. (2004) had concluded in their work that the concept of Windex works well with wet microburst situations. Nevertheless, the exact values used here for demarcation do not have much effect on analysis because this is done to explore the situations where INSAT-3DR Windex and RAOB Windex are referring to different weather events. The match of the exact numerical value of Windex from two different observations will not affect much the outcome till their values are in the range where they indicate similar weather events. Thus, to analyse these confusion matrices are generated using the different categories of Windex.

3 Results and discussion

In this section, the results are discussed in detail. Before discussing various results, it is necessary to evaluate the sensitivity of Windex on temperature and moisture profile. Few studies have already been conducted to evaluate the accuracies of atmospheric soundings from INSAT-3D/3DR sounders. Tarkeshwar et al. (2017) carried out a validation exercise of INSAT-3D sounder-derived profiles using ROABs, and Giri et al. (2022) carried out the validation of INSAT-3DR temperature profiles using GPS radiosonde and AIRS observations. Of late Gopikrishnan et al. (2023) performed an extensive evaluation of INSAT-3D and INSAT-3DR

temperature profile retrievals using in-situ observations and reanalysis data sets. All these studies suggested that errors in retrieved profiles of INSAT-3D/3DR are within standard limits. However, it is always recommended to perform a sensitivity of the parameter concerning the standard error ranges. It helps us to understand the error in parameters. For the sensitivity analysis of Windex, three experiments have been carried out. First, the standard tropical profile is taken. The temperature of each level is changed by 1, 2 and 3 °C and the change in the value of Windex is computed for change at each level shown in Fig. 2. The other parameters such as humidity remained fixed. The Fig. 2 shows that change in increase in temperature causes the increase in the value of temp. A similar exercise is repeated by changing humidity at each level by 10%, 20%, and 30% while the temperature remains constant represented in Fig. 3. In a final experiment, the combination of temperature and humidity changes are taken while computing the respective change in Windex shown in Fig. 4. The observations depicted in Figs. 3 and 4 also show similar behaviour. The change in Windex is highest for 3 °C change in temperature and 30% change in humidity. The value of Windex for sample tropical profile is approximately 38 and reaches to value of approximately 52 for the case of 3 °C and 30% change in temperature and humidity respectively. It shows that Windex is severely affected by the error in temperature and humidity particularly for larger errors.

First of all, the plots of Windex from the INSAT-3DR sounder and RAOBs for each station are shown in Fig. 5. The plots are drawn for each year from 2020–2022. For most of the stations, the values of Windex of INSAT-3DR data are similar to the values of Windex from RAOBs for the years 2020 and 2021. The figures show that the year 2021 shows the least difference and the year 2022 shows relatively larger

Fig. 2 Sensitivity of Windex on temperature

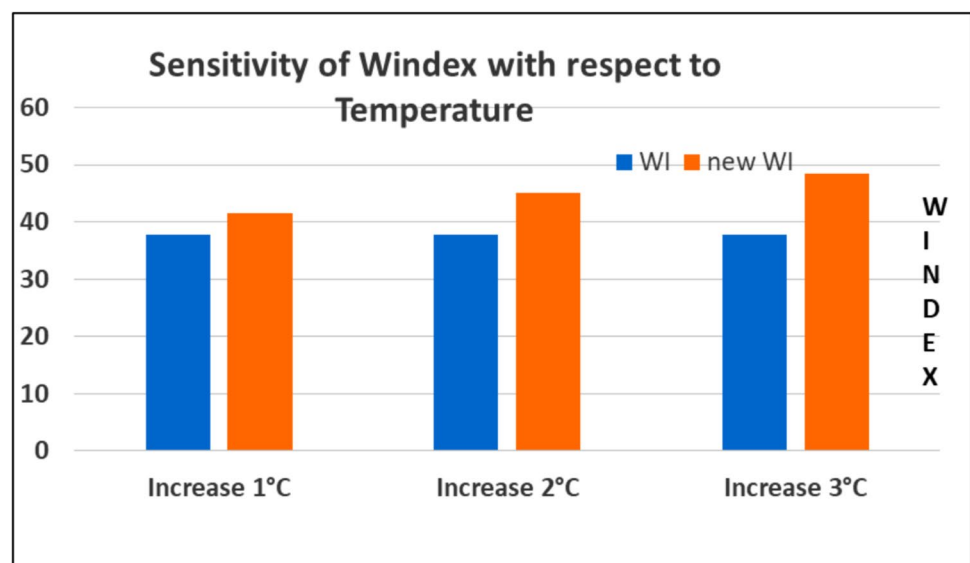
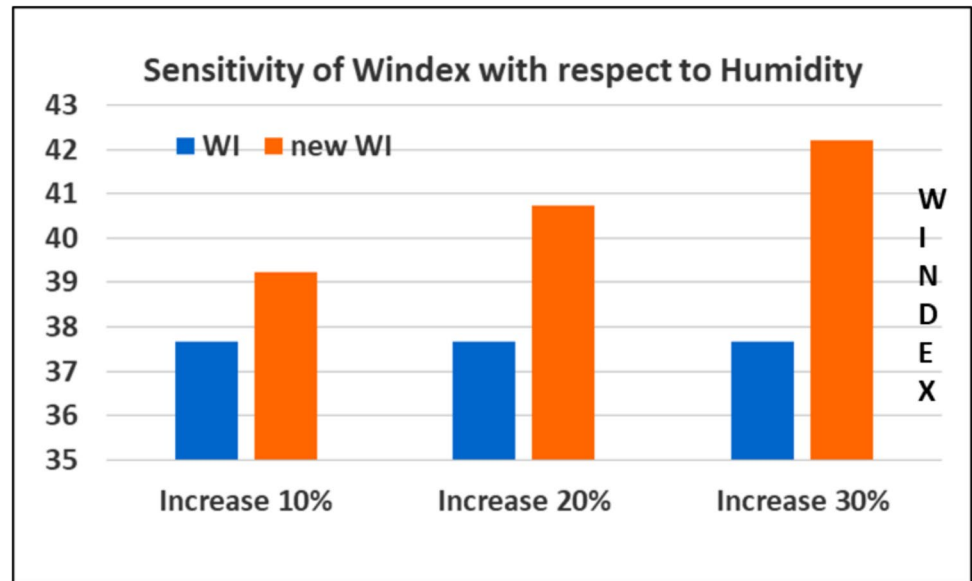
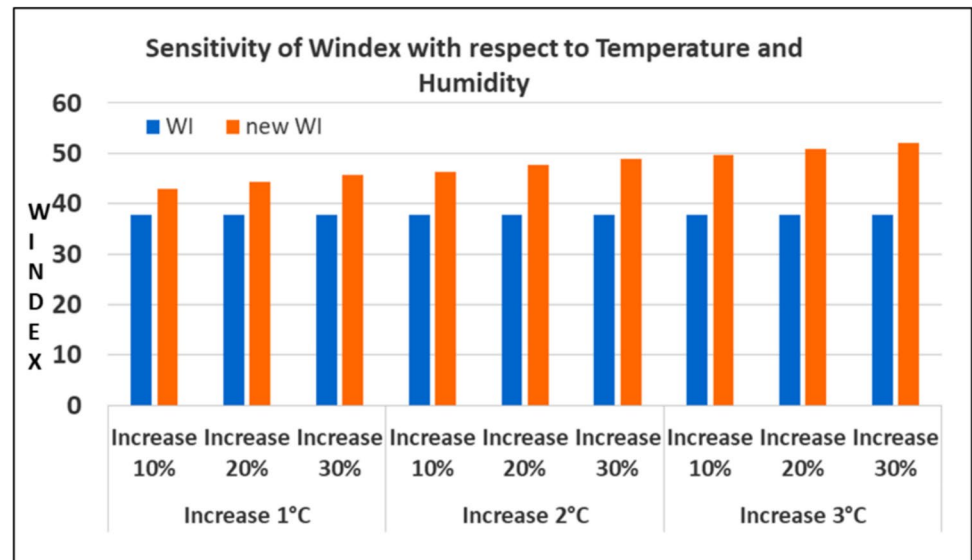


Fig. 3 Sensitivity of Windex on Humidity**Fig. 4** Sensitivity of Windex on temperature and moisture profile

differences in the Windex values. For year 2020 and 2021 the difference in INSAT WI and RAOB WI for different stations do not show any clear biases. Some stations show positive biases in INSAT WI whereas some stations show negative biases with respect to RAOB WI. Nevertheless, for year 2022, INSAT WI show negative bias. The negative bias may arise due to increased error in both temperature and humidity. These figures represent the holistic picture qualitatively. The quantitative assessments are shown in Figs. 6, 7, and 8.

Now, for more rigorous validation process is employed to assess the accuracy of Wind Index values derived from INSAT-3DR sounder data by comparing them to ground-based, in-situ data provided by the Indian Meteorological

Department (IMD) for the years 2020, 2021, and 2022. To carry out more quantitative validation exercise, several important statistical metrics were used to evaluate the performance of the Wind Index data. Correlation (CORREL) measures the degree of linear relationship between the predicted and observed Wind Index values. Figure 6 provides the validation results for Wind Index data across various weather stations. Positive values of CORREL (e.g., Thiruvananthapuram (0.49), Andaman and Nicobar Islands (0.28), Mangaluru (0.68), etc.) indicate a positive linear relationship, while a negative value (Karaikal (−0.49), Pondicherry (−0.49) etc.) suggests a negative correlation. Root Mean Squared Error (RMSE) quantifies the overall magnitude of errors between predicted and observed data, with lower

Fig. 5 Plots of Windex from INSAT-3DR sounder and RAOBs for 2020, 2021 and 2022

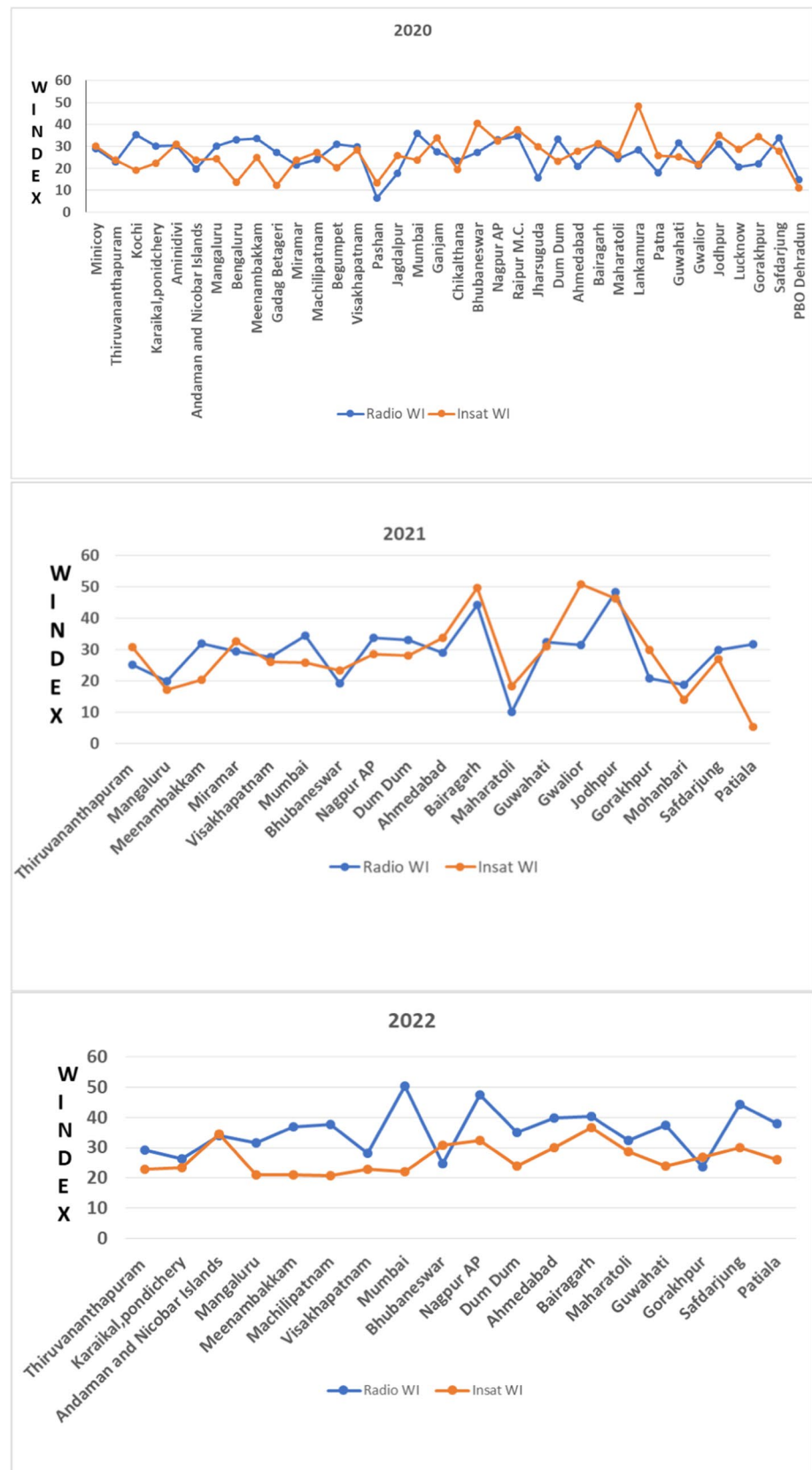
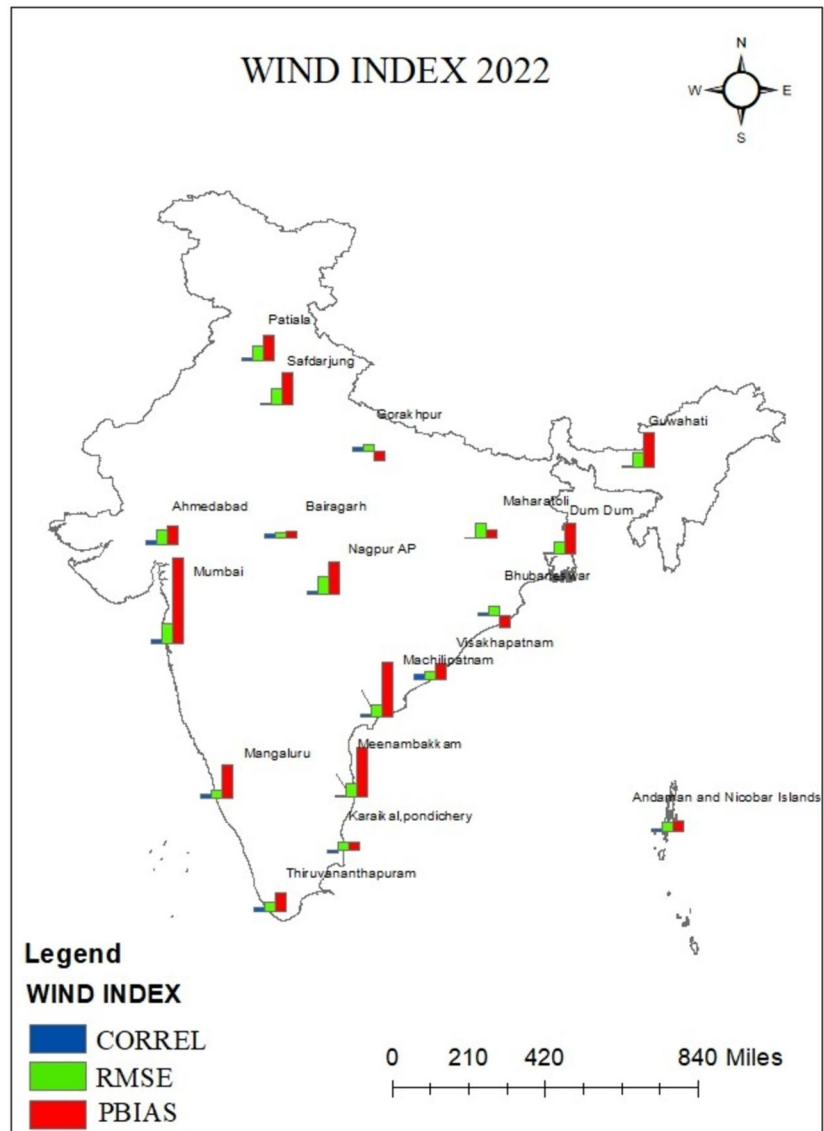


Fig. 6 Wind Index statics of 2022



values indicating better model performance. Thiruvananthapuram (12.68), despite a moderate correlation, has a relatively low RMSE, implying decent accuracy. Conversely, Meenambakkam (21.18) and Machilipatnam (17.86) exhibit high RMSE values, suggesting substantial errors in their Wind Index predictions. Percent Bias (PBIAS) gauges systematic over- or underestimation in predictions. Positive PBIAS values (e.g., Meenambakkam (74.4), Machilipatnam (82.2), and Mangaluru (50) signify overestimation, while negative values (e.g., Karaikal (−12.5), Pondicherry (−12.5), and Bhubaneswar (−19.2)) indicate underestimation. Lastly, the R-squared (R^2) values indicate how well the predicted values fit the observed data, with higher values indicating a better fit. Mangaluru demonstrates a strong fit with an R^2 of 0.68, indicating its data reliability.

Overall, among the stations, Mangaluru stands out with a strong correlation coefficient of 0.68 and a high R-squared

value of 0.68. This indicates a robust relationship between the derived Wind Index and the ground-based measurements. However, there is a significant bias of 50%, suggesting that the Wind Index may tend to overestimate wind conditions at this location. On the other hand, Meenambakkam shows a relatively low R-squared value of 0.03, indicating poor agreement between the derived and in-situ Wind Index data. The high bias of 74.4% and a large RMSE of 21.18 further underscore the lack of accuracy in this case.

Figure 7 presents a statistical analysis of Wind Index data for the year 2021 of various meteorological stations. Thiruvananthapuram is showing a relatively high correlation of 0.76, indicating a strong positive linear relationship between predicted and observed Wind Index values. The negative Mean Error (ME) of −3.49 suggests that, on average, the predictions underestimate the observed values, but the Root Mean Squared Error (RMSE) of 11.02 is relatively

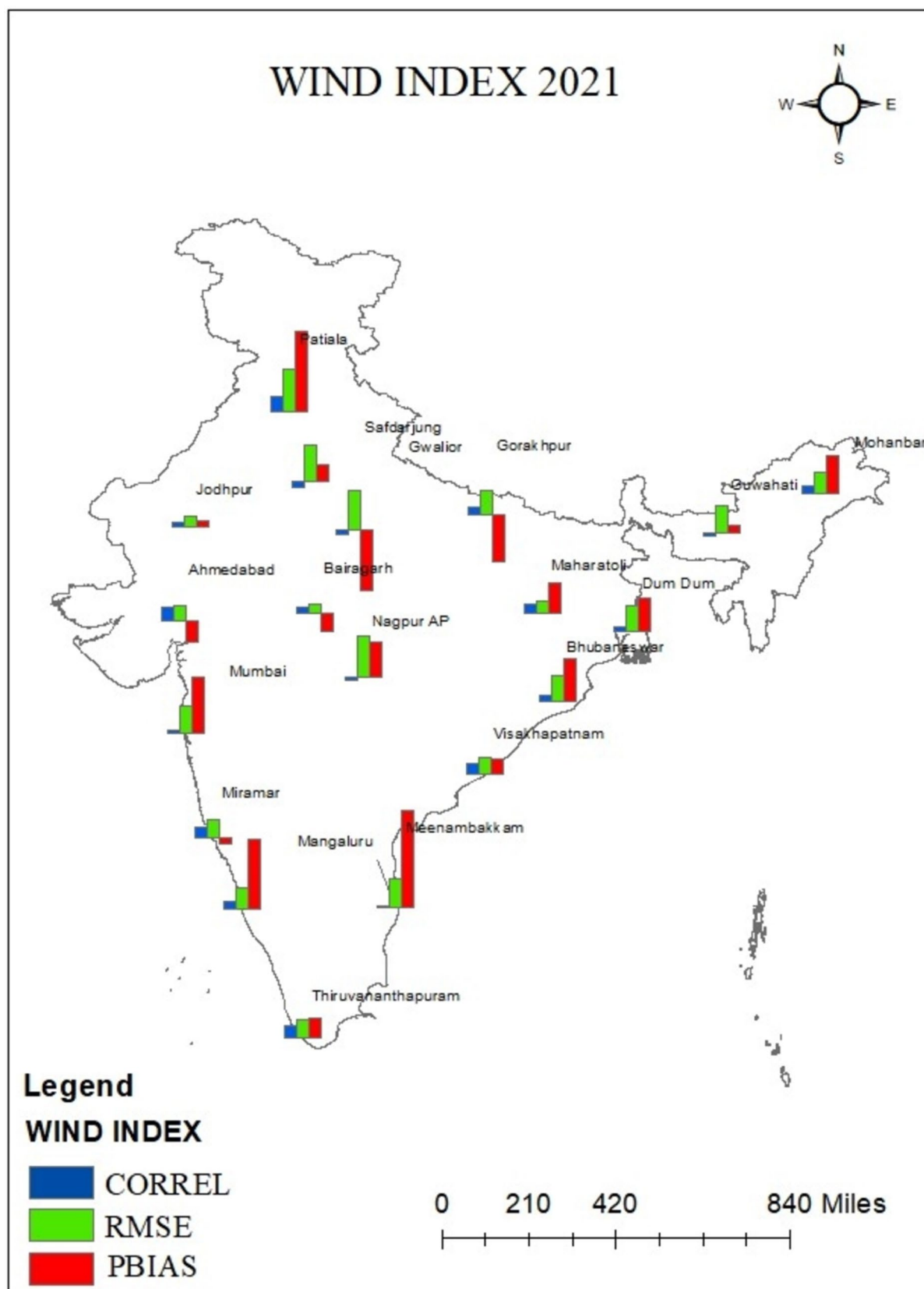


Fig. 7 Wind Index statics of 2021

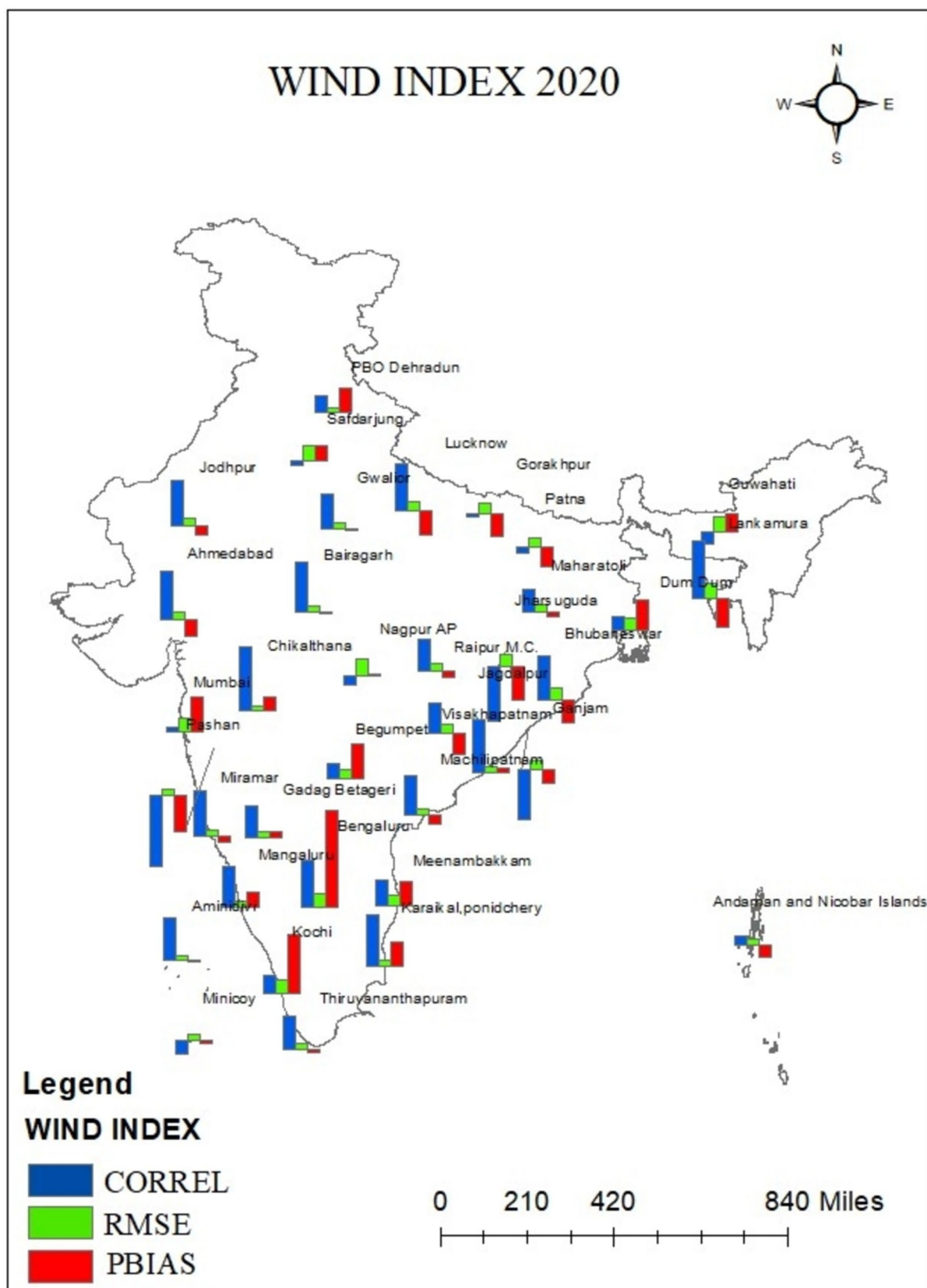


Fig. 8 Wind Index statics of 2020

low, indicating that the errors are not large. The Percent Bias (PBIAS) of 12% suggests a slight overestimation bias and a low R^2 value of 0.24.

On the other hand, Patiala stands out with a perfect correlation of 1.0, suggesting a flawless linear relationship between predicted and observed values. However, the extremely high ME and RMSE values indicate substantial errors in predicting Wind Index values, with a significant overestimation bias (PBIAS of 511.1%).

As shown in Fig. 8, in 2020, meteorological stations exhibited varying degrees of accuracy in Wind Index predictions. Thiruvananthapuram showed a moderate positive correlation (0.48) with observed values, a relatively low Mean Error (-0.87), and a moderate Root Mean Squared Error (RMSE) of 9.54. However, there was a minor underestimation bias (-3.6%), and the R^2 value was 0.23, indicating that the model captured only some of the observed data variability. In contrast, Kochi displayed weaker statistical results, with a correlation of 0.27, a high Mean Error of 16.22 (indicating significant overestimation), a substantial RMSE of 19.4, and a notably high Percent Bias of 84.5% (suggesting substantial overestimation bias). The R^2 value was only 0.07, signifying a poor fit between predicted and observed data. Most of the stations showed RMSE of daily insolation estimates within $3\text{--}5 \text{ MJm}^{-2}$ and correlation coefficient of 0.7–0.85 except for two hilly stations such as Kodaikanal and Shillong. It may be noted that the slopes of validation plots of one continental and one maritime station at New Delhi and Mumbai showed close to 1.0 (0.93 for New Delhi and 0.8 for Mumbai) with less bias (1.48 MJm^{-2} for New Delhi and 3.09 MJm^{-2} for Mumbai) in contrast to rest of the stations.

The validation exercise is further extended by making different comparisons in WMSI ranges. As it has already been discussed that WI is used to investigate and predict the high wind gusts scenarios. In such case, the different ranges of WI signify the different weather conditions. Windex value less than 10 shows convection microburst unlikely while Windex value between 10–49 shows wind gust should be less than 35KT. Therefore, the exact one on one comparison will not be able to represent its actual efficacy. For better understanding, it would be interesting to make comparison in different ranges.

Table 3 represents the confusion matrices for ranges of WI values provided in Table 2. The confusion matrices are computed for all the years 2020, 2021 and 2022.

From Table 3 it is evident that maximum number of points lie on the diagonal elements and the number of off diagonal points are relatively low. It shows that for maximum cases INSAT-3DR and RAOB represents the same weather conditions based on the WI values. This is true for all the years. The analysis shows that neither satellite nor RAOB show any very extreme case corresponding to value

Table 2 Convective Wind Gust Vs WMSI Values

Convective Wind Gust Magnitude Corresponding to WMSI Values	
WI	Wind Gusts (KT)
< 10	Convection Microburst Unlikely
10–49	< 35
49–79	35–49
> 79	> 50

Table 3 Windex cases in a range

	< 10	10–49	49–79	> 79
2022				
< 10	0	2	0	0
10–49	2	19	0	0
49–79	0	1	0	0
> 79	0	0	0	0
2021				
< 10	4	33	1	0
10–49	35	456	22	0
49–79	4	58	5	0
> 79	0	0	0	0
2020				
< 10	6	45	0	0
10–49	57	750	47	0
49–79	2	74	3	0
> 79	0	0	0	0

greater than 79. Table 3 also suggests that the likelihood of maximum error attributes to the < 10 and 10–49 ranges. In 2022 19 cases were found in the range 10–49. In 2021 456 cases were found in this range and 2020 total number of cases was 750. This errors in WI of values of INSAT-3DR may result in omission or commission error in wind gusts error.

The spatial plot made for a mean of years 2022, 2021, and 2020 is shown in Fig. 9.

4 Case 1: Delhi Thunderstorm event on 28th May 2023 evening

Predicting in advance the abrupt convective development and the squally weather it brings is extremely challenging. On the other hand, it is discovered that the almost real-time data obtained from Radiosonde and its by-products is highly helpful in tracking and documenting the history of this abrupt change. At 1500 IST on May 28, 2023, the Wind Index (WI) values (> 45 knots) show that the weather is favourable and that there is a possibility of rain and squally

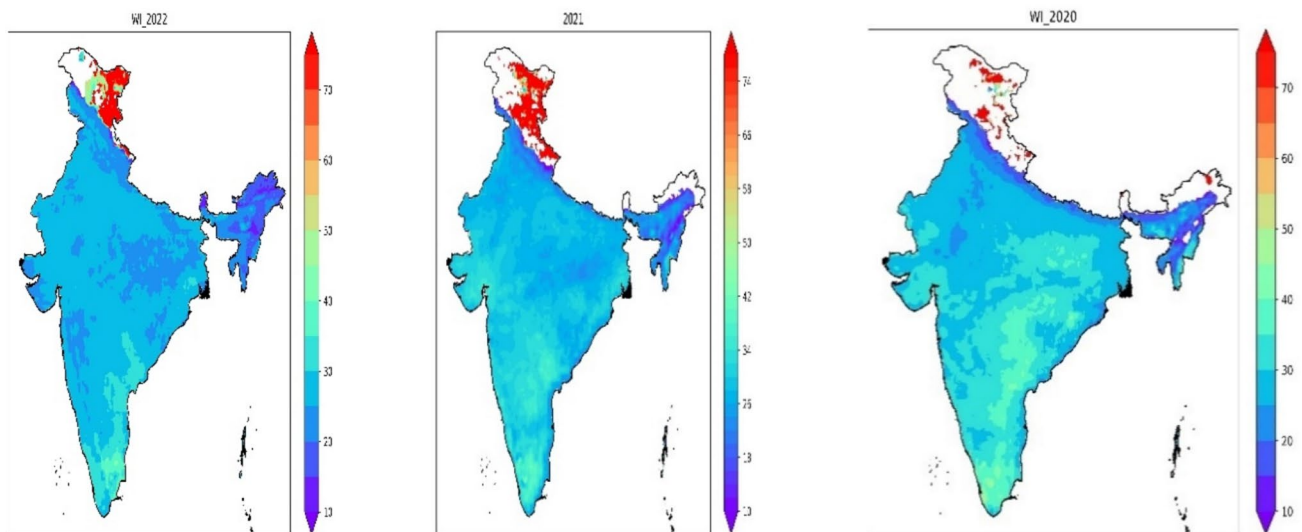


Fig. 9 Spatial mean plot of Wind Index 2022,2021 and 2020

winds. However, a comparable kind of investigation across a bigger area with a greater number of cases with additional information on wind confluence and moisture availability, etc., is required to evaluate the current casting potential of WI.

5 Case 2: Helicopter suspension event on 27th June 2023

Partly cloudy weather with thundershowers in Patna airport on the 27th of June at 1500 IST wind gusts were reported WINDEX value greater than 30, it is noted a wind gust negatively impacts the safety and causes the helicopter to move like an externally suspended aircraft, Windex high capability to work as an indicator of extreme events with Windex value will become a good forecasting tool for forecaster to make early decision of helicopter take-off. Various case studies were done and found that most suspension cases occur whenever the Windex value is more than 30.

6 Conclusion

This paper presents a measure Windex, has a potential to predict microburst, is derived from INSAT-3DR temperature and humidity profiles. Windex incorporates sounding aspects such as low-level moisture and sub-cloud lapse rate. A preliminary validation exercise of wind index data is done for INSAT 3DR and Radiosonde data over the Indian region. Carried out for 2022, 2021 and 2020 for 38 stations distributed over Indian land mass. In this study, we found that

a high WINDEX assessment stimulates significant vertical convective formation and has the potential to increase damage over the region. The accuracy of the product depends on the accuracy of the retrieved temperature and humidity profile. It is noted that some sites cannot provide comparison results due to empty sample points. Geostationary satellite has some limitation in cloudy condition its unable to penetrate infra-red rays hence any weather event indication we can get it prior to 3–4 h of the event. This indices Windex has proven to be a tool for initial assessment and evaluation of thunderstorm events. The method ensures a high capability to work as an indicator of thunderstorm events and have worldwide coverage. To make the method more reliable and accurate more case analysis is required. Further analysis is required for making such comparison for more robust conclusions. The analysis will use more satellite indices to design a new model for thunderstorm detection trends and forecasting.

Acknowledgements The authors are very much grateful and rendered by the Director General of Meteorology of India Meteorological Department (IMD) Dr. M. Mohapatra for facilitating this research work. The authors are deeply thankful to the Director, Space Applications Centre (SAC) and Deputy Directors, EPSA and SIPA, SAC for their continuous encouragement during work. The authors are sincerely grateful to Mr. Sai Krishnan for his guidance and support. The work was carried out under the ISRO-IMD collaboration on Multi-mission Data Reception and Processing System (MMDRPS).

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D: A.k. Mitra

E:R.K. Giri,

A and B: took the lead in writing the manuscript.

E:helped in designing and Planning the methodology to reach the conclusion.

B, C & A:designed the model and the computational framework
D and E analyzed the data.

All authors provided critical feedback and helped to shape the research, analysis, and manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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